Number Sequences (Set B)

Year 4 · **Mark Scheme**

Lesson Worksheet — Answers

Spotting the Pattern

1. 15 — gap +3 throughout (3, 6, 9, 12, 15).
2. 30 — gap -5 throughout (50, 45, 40, 35, 30).
3. D (32) — each term doubles (2, 4, 8, 16, 32).
4. A, C, E — these all increase by exactly 4 each time; B increases by 4,4,5 and D increases by 4,5,4, so they are inconsistent.
5. 35 — gap +7 throughout (7, 14, 21, 28, 35).
6. C (60) — gap -10 throughout (100, 90, 80, 70, 60).
7. 36 — square numbers: 1^2, 2^2, 3^2, 4^2, 5^2, 6^2 = 1, 4, 9, 16, 25, 36.
8. 15 — 5, 10, 15, 20, 25 (gap +5 throughout).
9. D (32) — each term doubles (1, 2, 4, 8, 16, 32).
10. A, C, D — each term is multiplied by 3 (geometric), so the next term is 81×3=243; it is not arithmetic, and the increase is not a constant 6 (it increases by 6, 18, 54).
11. 14 — gap +3 throughout (2, 5, 8, 11, 14).
12. C (13) — Fibonacci rule: 5+8=13.
13. 36 — gap -6 throughout (60, 54, 48, 42, 36).
14. 16 — each term doubles: 2, 4, 8, 16, 32.
15. A, B, D — these are the first five square numbers, the next term is 36 (6^2), and the gap between terms increases each time (3, 5, 7, 9); the sequence does not increase by a constant amount, and it is increasing, not decreasing, so C and E are false.

Applying the Pattern

16. 39 — gaps increase by 2 each time: 4→7(+3), 7→12(+5), 12→19(+7), 19→28(+9), 28→39(+11).
17. 6.25 — each term is halved: 100, 50, 25, 12.5, 6.25.
18. 21 — Fibonacci-style: 8+13=21.
19. 17 — 3→5 (3×2−1=5), 5→9 (5×2−1=9), 9→17 (9×2−1=17).
20. A, C, E — each term is the sum of the two preceding terms; B and D increase by a constant amount instead (arithmetic, not Fibonacci-style).
21. 50 — gaps increase by 3 each time: 5→8(+3), 8→14(+6), 14→23(+9), 23→35(+12), 35→50(+15).
22. 1, 5, 13, 29 — term1=1; term2=1×2+3=5; term3=5×2+3=13; term4=13×2+3=29.
23. 21 — triangular numbers: each term adds the next counting number (1, +2=3, +3=6, +4=10, +5=15, +6=21).
24. The gap increases by 2 each time (+3, +5, +7, +9); the next term is 27+9=36.
25. A, C, E — in each of these, the gaps themselves increase by a constant amount; B has a constant gap (not increasing), and D's gaps double rather than increase by a fixed amount.
26. 12.5 — each term is halved: 200, 100, 50, 25, 12.5.
27. 7, 12, 19 — 2+5=7, 5+7=12, 7+12=19.
28. 720 — each term is multiplied by an increasing amount: 1×2=2, 2×3=6, 6×4=24, 24×5=120, 120×6=720.
29. A, C, D — each has a constant multiplying ratio (×2, ×0.5, ×3 respectively); B increases by a constant amount (arithmetic, not geometric) and E also increases by a constant amount (+2 each time).
30. 16, 28 — 10−2=8, 8×2=16; 16−2=14, 14×2=28. Sequences in Context

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Sequences in Context

31. 47 — the rule is "double the previous term, then add 1": 2×2+1=5, 5×2+1=11, 11×2+1=23, 23×2+1=47.

32. 2, 5, 11, 23 — 2×2+1=5, 5×2+1=11, 11×2+1=23.

33. C) 8 — each term is the sum of the two before it: 3+5=8.

34. 15 — triangular numbers: 1, 3, 6, 10, 15 (add one more each time: +2, +3, +4, +5).

35. A, C, D — each term is multiplied by 2 (A), so the next term is 24×2=48 (C), making this a geometric sequence (D). It increases by a changing amount, not a constant 3, so B is false; it is not arithmetic, so E is false.

36. 10 — term2 = 4×2−2=6; term3 = 6×2−2=10.

37. C) 22 — the gaps are +2, +3, +4, +5, so the next gap is +6, giving 16+6=22.

38. 3, 5, 8 — 1+2=3, 2+3=5, 3+5=8.

39. 13 — alternating +4 and −2: 5+4=9, 9−2=7, 7+4=11, 11−2=9, 9+4=13.

40. The sequence is NOT arithmetic because the gaps between terms are not constant (2, 4, 8 — they keep growing). It is a geometric sequence instead, because each term is found by multiplying the previous term by the same constant ratio (×2).

41. 80 — each term is doubled: 5→10→20→40→80.

42. 5, 8, 13 — 2+3=5, 3+5=8, 5+8=13.

43. B) 14 — subtracting 3 from 10 gives 7, then multiplying by 2 gives 14.

44. A, C, D — each of these has a constant multiplying ratio: A (×2), C (×0.5), D (×2). B and E increase by a constant amount instead (arithmetic, not geometric).

45. 10, 28 — (1+4)×2=10, then (10+4)×2=28.

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Homework Worksheet — Answers

Quick Practice

1. 25 — gap +5 throughout (5, 10, 15, 20, 25).

2. 50 — gap -10 throughout (90, 80, 70, 60, 50).

3. D (48) — each term doubles (3, 6, 12, 24, 48).

4. A, C, E — these increase by exactly 3 each time; B increases by 3,4,3 and D increases by 3,2,3, so they are inconsistent.

5. 40 — gap +8 throughout (8, 16, 24, 32, 40).

6. B (45) — gap -9 throughout (81, 72, 63, 54, 45).

7. 125 — cube numbers: 1^3, 2^3, 3^3, 4^3, 5^3 = 1, 8, 27, 64, 125.

8. 12 — 4, 8, 12, 16, 20 (gap +4 throughout).

9. D (96) — each term doubles (3, 6, 12, 24, 48, 96).

10. A, C, D — each term is multiplied by 3 (geometric), so the next term is 54×3=162; it is not arithmetic, and the increase is not constant (it increases by 4, 12, 36).

11. 24 — gap +5 throughout (4, 9, 14, 19, 24).

12. D (26) — Fibonacci rule: 10+16=26.

13. 42 — gap -7 throughout (70, 63, 56, 49, 42).

14. 24 — each term doubles: 3, 6, 12, 24, 48.

15. A, B, C — these are even numbers, the next term is 12, and each term increases by a constant amount of 2; the gap does not increase (it stays at 2), and the sequence is increasing, not decreasing, so D and E are false.

Mixed Practice

16. 33 — gaps increase by 2 each time: 3→5(+2), 5→9(+4), 9→15(+6), 15→23(+8), 23→33(+10).

17. 2 — each term is divided by 3: 162, 54, 18, 6, 2.

18. 23 — Fibonacci-style: 9+14=23.

19. 30 — 2→6 (2×2+2=6), 6→14 (6×2+2=14), 14→30 (14×2+2=30).

20. A, C, D — each term is the sum of the two preceding terms; B and E increase/multiply by a constant amount instead (not Fibonacci-style).

21. 51 — gaps increase by 3 each time: 6→9(+3), 9→15(+6), 15→24(+9), 24→36(+12), 36→51(+15).

22. 2, 4, 10, 28 — term1=2; term2=2×3−2=4; term3=4×3−2=10; term4=10×3−2=28.

23. 28 — triangular numbers: 1,3,6,10,15,21,28 (adding 1,2,3,4,5,6,7 in turn).

24. The gap increases by 2 each time (+3, +5, +7, +9); the next term is 26+9=35.

25. A, C, E — in each, the gaps themselves increase by a constant amount; B has a constant gap (not increasing) and D's gaps multiply by 3 rather than increasing by a fixed amount.

26. 5 — each term is divided by 3: 405, 135, 45, 15, 5.

27. 7, 11, 18 — 3+4=7, 4+7=11, 7+11=18.

28. 360 — each term is multiplied by an increasing amount: 1×3=3, 3×4=12, 12×5=60, 60×6=360.

29. A, C, D — each has a constant multiplying ratio (×2, ×0.5, ×3 respectively); B and E increase by a constant amount (arithmetic, not geometric).

30. 8, 22 — starting at 1: add 3 → 4, multiply by 2 → 8 (first new term); then add 3 → 11, multiply by 2 → 22 (second new term).

Standard / Advanced

31. 8 — doubling 3 gives 6, then adding 2 gives 8.

32. 29 — the gap is +5 throughout; the next term after 19 is 24, and the term after that is 29.

33. B) 14 — subtracting 2 from 9 gives 7, then doubling gives 14.

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34. 15 — the gap is −7 throughout; the next term after 29 is 22, and the term after that is 15.

35. 10 — adding 3 to 2 gives 5, then doubling gives 10.

36. 3, 5, 9, 17 — 3×2−1=5, 5×2−1=9, 9×2−1=17.

37. D) 21 — each term is the sum of the two before it: 8+13=21.

38. 21 — triangular numbers: 1, 3, 6, 10, 15, 21 (add one more each time: +2, +3, +4, +5, +6).

39. A, C, D — each term is multiplied by 2 (A), so the next term is 32×2=64 (C), making this geometric (D). It does not increase by a constant amount, so B is false, and it is not arithmetic, so E is false.

40. 11 — term2 = 5×2−3=7; term3 = 7×2−3=11.

41. C) 23 — the gaps are +2, +3, +4, +5, so the next gap is +6, giving 17+6=23.

42. 4, 7, 11 — 1+3=4, 3+4=7, 4+7=11.

43. 14 — alternating +4 and −2: 6+4=10, 10−2=8, 8+4=12, 12−2=10, 10+4=14.

44. A, C, E — each has a constant multiplying ratio: A (×2), C (×0.5), E (×4). B and D increase by a constant amount instead, so they are arithmetic, not geometric.

45. 10, 24 — (3+2)×2=10, then (10+2)×2=24.

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